

EXPERIMENT NO. – 10

Code:

```
import numpy as np

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense


# Input Data (Days)
X = np.array([[1], [2], [3], [4], [5]], dtype=float)


# Output Data (Sales)
y = np.array([[100], [120], [130], [150], [170]], dtype=float)


# Normalize data
X = X / 5
y = y / 200


# Create Deep Learning Model
model = Sequential()


# Hidden Layer
model.add(Dense(8, input_dim=1, activation='relu'))


# Output Layer
model.add(Dense(1, activation='linear'))


# Compile Model
model.compile(optimizer='adam',
              loss='mean_squared_error')


# Train Model using Backpropagation
```

```
model.fit(X, y, epochs=500, verbose=0)
```

```
# Predict Future Sales
```

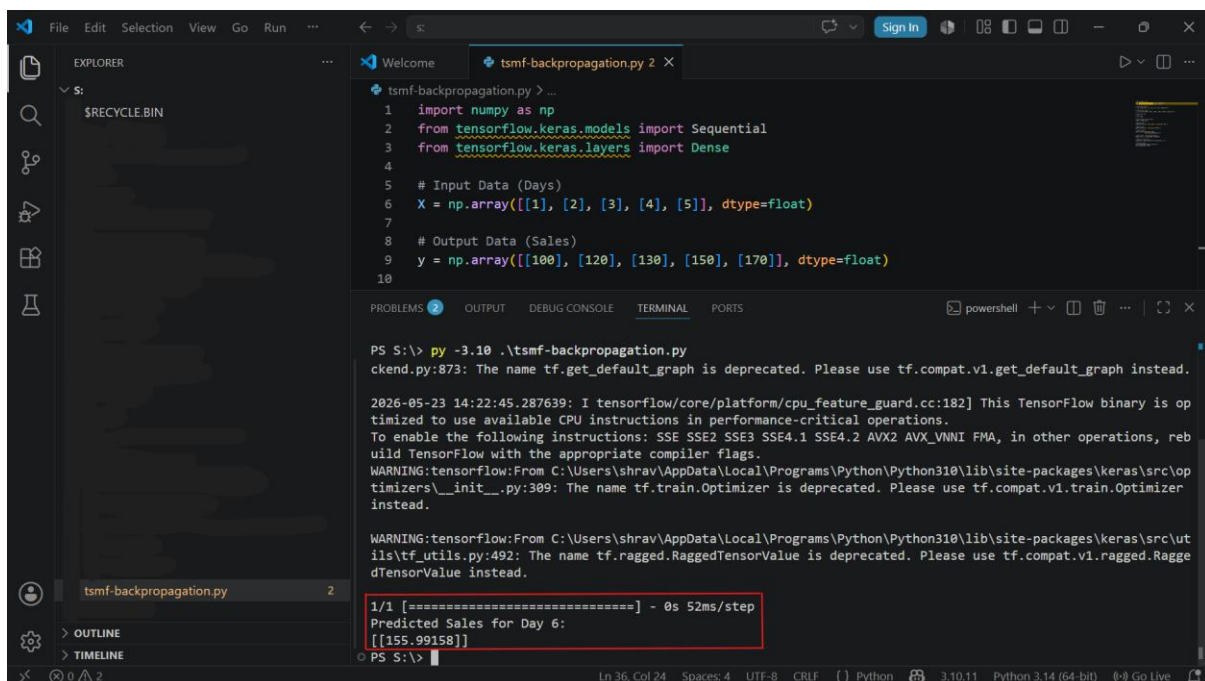
```
prediction = model.predict(np.array([[6]]) / 5)
```

```
# Display Output
```

```
print("Predicted Sales for Day 6:")
```

```
print(prediction * 200)
```

Output:



The screenshot shows a Visual Studio Code editor with a Python file named `tsmf-backpropagation.py` open. The code defines input and output data for a neural network model and prints the predicted sales for day 6. The terminal output shows the execution of the script, including several TensorFlow warnings and the final prediction result.

```
1 import numpy as np
2 from tensorflow.keras.models import Sequential
3 from tensorflow.keras.layers import Dense
4
5 # Input Data (Days)
6 X = np.array([[1], [2], [3], [4], [5]], dtype=float)
7
8 # Output Data (Sales)
9 y = np.array([[100], [120], [130], [150], [170]], dtype=float)
10
```

Terminal Output:

```
PS S:\> py -3.10 .\tsmf-backpropagation.py
ckend.py:873: The name tf.get_default_graph is deprecated. Please use tf.compat.v1.get_default_graph instead.

2026-05-23 14:22:45.287639: I tensorflow/core/platform/cpu_feature_guard.cc:182] This TensorFlow binary is op
timized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE SSE2 SSE3 SSE4.1 SSE4.2 AVX2 AVX_VNNI FMA, in other operations, reb
uild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:From C:\Users\shrav\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\op
timizers\__init__.py:309: The name tf.train.Optimizer is deprecated. Please use tf.compat.v1.train.Optimizer
instead.

WARNING:tensorflow:From C:\Users\shrav\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\ut
ils\tf_utils.py:492: The name tf.ragged.RaggedTensorValue is deprecated. Please use tf.compat.v1.ragged.Ragge
dTensorValue instead.

1/1 [=====] - 0s 52ms/step
Predicted Sales for Day 6:
[[155.99158]]
```